

RSPC: A Benchmark for Modeling Stress and Psychiatric Conditions in Digitally Mediated Relationships using Psychiatrist Annotations

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Abstract

In NLP, mental health conditions are often modeled as isolated phenomena, without interpersonal context. We use Reddit posts about long-distance relationships to capture both mental health distress and associated relational triggers. We introduce the Relational Stress and Psychiatry Corpus (RSPC), containing 1,799 Reddit posts annotated by psychiatrists for diagnostic categories, including the most prevalent mood disorders (anxiety and depression), relational stressor triggers, and indications of relationship phase. We benchmark seven fine-tuned transformer models and five large language models across the tasks of multi-label disorder classification, relational trigger detection, and temporal phase prediction. We find clear task-dependent differences between model families, with Claude-3-Haiku achieving the best disorder classification performance (Macro-F1 = 0.538) and GPT-4o obtaining the strongest relational trigger detection performance (Macro-F1 = 0.519), suggesting distinct model capabilities. We further find strong associations between anxiety disorders and chronic relational uncertainty. Overall, RSPC establishes a benchmark for NLP tasks that consider relational context and supports a shift from individual-centric to context-aware mental health modeling, capturing the social and temporal dynamics of distress.

1 Introduction

As global mobility increases, digital platforms mediate intimate relationships, profoundly impacting attachment and separation. However, computational mental health predominantly models psychiatric distress as an individual-level phenomenon

(treating symptoms as isolated internal states independent of external social triggers). This ignores the fact that relational context is vital: symptoms emerge from recurring stressors such as communication silences, jealousy, commitment ambiguity, and reunion-separation cycles. We must therefore shift toward relationally grounded natural language processing frameworks. Clinical studies firmly associate relationship-mediated stress with severe anxiety, adjustment difficulties, clinical depression, and insomnia (Stafford, 2004; Neustaedter and Greenberg, 2012).

Prior datasets largely fail to capture this complexity, reducing distress to coarse binary classification (Coppersmith et al., 2015; Losada et al., 2018; Yates et al., 2017; Cohan et al., 2018). They ignore the diverse mood and anxiety subtypes defined in DSM-5-TR, relying heavily on user-level annotations that strip away surrounding relational context. Furthermore, these benchmarks consistently operationalize temporality merely as chronological time (modeling time strictly via linear posting histories or absolute timestamps). This mechanistic approach entirely misses the underlying developmental phases of a relationship, preventing models from detecting how specific temporal transitions trigger escalating psychiatric symptoms over extended observational periods.

To address this, we focus on long-distance relationships. LDRs exclusively rely on digital communication, eliminating unrecorded physical interactions and providing an ideal context for observing relationally mediated distress. Unlike prior datasets that used coarse MTurk crowdsourcing, we collaborated with four psychiatrists from

Andhra University to develop a schema aligned with the DSM-5-TR (Association, 2022) and ICD-11 (World Health Organization et al., 2018) criteria. Four trained annotators independently evaluated each post for symptoms, stressors, and temporal phases. Disagreements were adjudicated strictly, yielding substantial inter-annotator agreement and high reliability across all evaluated clinically informed annotation tiers.

Leveraging this framework, we introduce the **Relational Stress and Psychiatry Corpus (RSPC)**, comprising 1,799 Reddit posts from long-distance communities annotated for DSM-5-TR/ICD-11-aligned symptoms, relational stressors, and temporal phases. Our contributions are threefold: (1) we introduce the first benchmark linking clinically grounded psychiatric categories with relational stressors and temporal phases in digitally mediated relationships, (2) we benchmark transformers and LLMs across disorder classification, trigger detection, and temporal reasoning tasks, and (3) we demonstrate that relational context provides a measurable signal for psychiatric inference within communication environments.

2 Related Work

Recent work on **Mental Health Detection in Social Media** has primarily focused on detecting depression, suicidality, and anxiety from social media language (Coppersmith et al., 2015; Yates et al., 2017; Cohan et al., 2018; Losada et al., 2018). Earlier approaches relied on lexical features, whereas recent work has adopted transformer architectures (Nadeem, 2016; Harrigian et al., 2021). Large language models have further expanded the scope of zero-shot psychiatric inference. Existing benchmarks, however, largely model psychiatric distress as an individual-level attribute detached from interpersonal context. Relational dynamics such as communication disruption, attachment insecurity, and jealousy remain underrepresented despite their clinical relevance.

Relational and Contextual Modeling. Prior work has explored conversational dynamics (Alghowinem et al., 2016), online social support (De Choudhury et al., 2013), emotional disclosure (Pavlova and Berkers, 2020), and longitudinal behavioral change (MacAvaney et al., 2018; Sekulić et al., 2018). While recent studies extract life events and social contexts to explain depression (Chen et al., 2024; Lv et al., 2025), these

approaches still largely emphasize broader environmental factors over clinically meaningful interpersonal stressors, such as communication gaps or commitment ambiguity. No prior work explicitly links event-contingent relational stressors with DSM-5-TR psychiatric symptom categories.

Clinical Grounding and Ethics. Recent studies highlight limitations in clinical grounding, including noisy self-disclosure labels, inconsistent annotations, and weak alignment with diagnostic frameworks (Chancellor et al., 2019; Harrigian et al., 2021). Researchers increasingly advocate for clinically informed annotation protocols and collaboration with mental health professionals (Low et al., 2020; Zirikly et al., 2019). Prior datasets, however, predominantly rely on coarse binary depression labels rather than differentiated symptom-level psychiatric categories in relational settings.

Large Language Models for Mental Health. Recent work explores LLM prompting for psychiatric classification, counseling simulation, and risk detection (Guo et al., 2024; Gilardi et al., 2023). Prompted LLMs often perform strongly on socially contextual tasks, but their behavior under clinically grounded multi-label psychiatric inference remains underexplored. Existing work also lacks a systematic comparison between fine-tuned transformers and prompted LLMs for relationally grounded psychiatric reasoning.

Research Gap & Positioning of RSPC: Existing mental health NLP research predominantly models psychiatric distress as an individual-level phenomenon using coarse diagnostic labels, with limited consideration of the relational and interpersonal dynamics through which symptoms emerge. Prior work on social-context modeling has focused primarily on generic interaction patterns or longitudinal behavior, while clinically meaningful relational stressors, such as communication disruption, attachment insecurity, and commitment ambiguity, remain underexplored. Furthermore, existing benchmarks lack DSM-5-TR-aligned multi-label psychiatric annotations and systematic evaluation of LLM-based relational psychiatric reasoning. RSPC addresses these gaps through clinically grounded relational-context modeling and comprehensive benchmarking of transformers and LLMs for psychiatric symptom inference.

3 RSPC CONSTRUCTION

This section presents the methodology and data-collection pipeline used to construct RSPC as described in Fig. 1.

3.1 Dataset Collection & Construction

We introduce the **Relational Stress and Psychiatry Corpus (RSPC)**, a benchmark for studying the expression of psychiatric symptoms in digitally mediated romantic relationships. The dataset comprises publicly available Reddit posts from long-distance relationship communities (r/LongDistance, r/LDR) collected between January 2020 and December 2023 and filtered for narrative completeness, relational relevance, and linguistic consistency. All usernames, personal identifiers, and proper nouns were anonymized using placeholder tokens (e.g., [USER], [PLACE]). Consistent with prior mental health NLP research, only publicly accessible posts were used, with no attempts made to infer user identities or contact individuals directly.

3.2 Data Annotation & Guidelines

The annotation framework was developed in consultation with a team of four licensed psychiatrists from Andhra University and grounded in DSM-5-TR (Association, 2022) and ICD-11 (World Health Organization et al., 2018) diagnostic criteria. Each Reddit post was independently annotated by four trained annotators using a clinically informed coding manual, with disagreements resolved through adjudication. Inter-rater reliability, measured using Cohen’s κ , demonstrated substantial agreement across all annotation tiers, including psychiatric symptoms (0.78), relational stressors (0.72), and temporal relationship phases (0.81).

1. **Tier A: Psychiatric Symptom Categories.** Posts were annotated for five clinically grounded symptom categories: Major Depressive Disorder (MDD), Generalized Anxiety Disorder (GAD), Separation Anxiety Disorder (SAD), Adjustment Disorder (ADJ), and Insomnia, using DSM-5-TR/ICD-11-aligned criteria adapted for textual inference.
2. **Tier B: Relational Stressor Triggers.** Posts were annotated for relational stressors, including Commitment Ambiguity, Lack of Communication, Reunion/Separation Stress, Trust/Fidelity Issues, Jealousy/Insecurity, Silence

Table 1: Psychiatric symptom label prevalence across splits.

Split	<i>n</i>	ADJ	GAD	SAD	MDD
Train	1,259	74.7%	71.3%	42.7%	17.1%
Validation	179	77.1%	69.8%	48.6%	16.8%
Test	361	72.9%	70.9%	43.8%	16.9%
Full	1,799	74.5%	71.1%	43.5%	17.0%

Note: Insomnia prevalence is 0.9%, 1.1%, 2.2%, and 1.2% for Train, Validation, Test, and Full, respectively.

Table 2: Relational trigger prevalence across dataset splits (%).

Split	CA	LC	RS	TF	J
Train	67.0	60.4	21.5	17.7	18.0
Validation	67.0	66.5	21.8	22.9	15.6
Test	62.9	64.0	21.9	18.0	19.4
Full	66.2	61.7	21.6	18.3	18.0

Note: Full-corpus prevalence for the less frequent triggers SG, SMS, and TZM is 5.7%, 2.1%, and 1.6%, respectively.

Gaps, Social Media Surveillance, and Time-zone Misalignment.

3. **Tier C: Temporal Relationship Phase.** Posts were assigned to one of four temporal phases: SEPARATION, ANTICIPATION, REUNION, or UNKNOWN, modeling event-contingent rather than chronological time.

3.3 Dataset Statistics

The final corpus contains 1,799 Reddit posts, split into train/validation/test partitions with a 70:10:20 stratified ratio. Adjustment Disorder (74.5%) and GAD (71.1%) are the dominant psychiatric categories, while MDD (17.1%) and Insomnia (1.2%) remain sparse. Commitment Ambiguity (66.2%) and Lack of Communication (61.7%) are the most common relational stressors. Temporal phase labels are dominated by SEPARATION (65.1%). Table 1 and Table 2 report full per-split label distributions; co-occurrence statistics are provided in Appendix A.

3.4 Inter-annotator Agreement Scores

Inter-annotator agreement was evaluated using Cohen’s κ across all annotation tiers, demonstrating substantial agreement for psychiatric symptoms, relational triggers, and temporal phase annotations. A detailed analysis of the agreement and tier-wise scores is provided in Appendix 7.

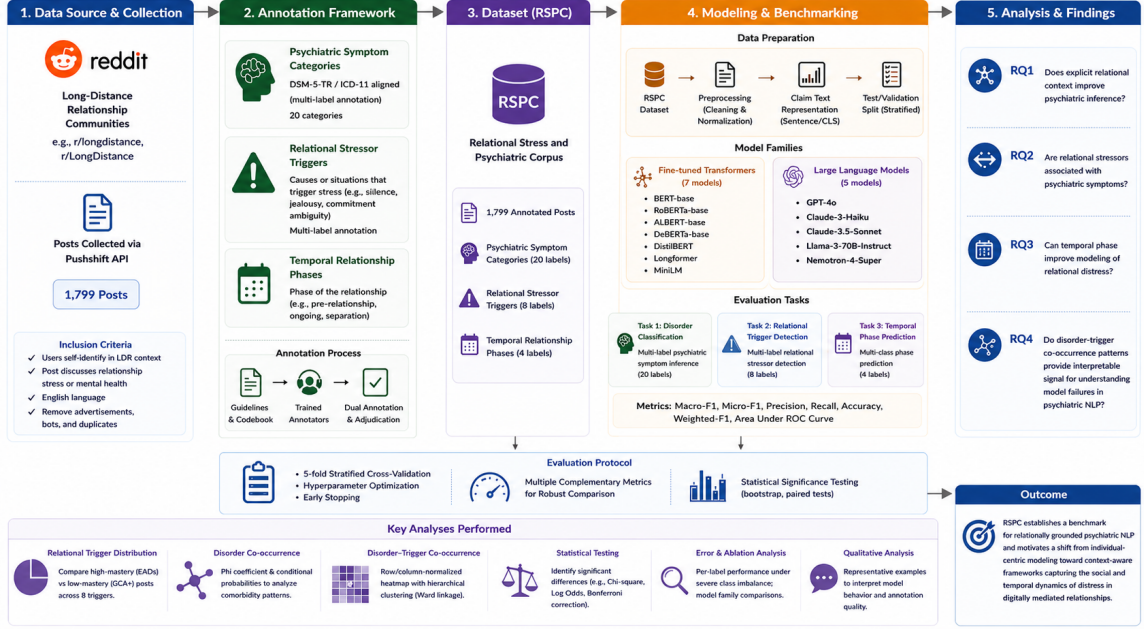


Figure 1: Workflow diagram of RSPC

4 Methodology

This section presents the major experiments conducted on **RSPC**, involving LLM- and PLM-based approaches. Detailed experimental settings and results are described in the following subsections.

4.1 Models and Experimental Setup

We evaluate RSPC across three benchmark tasks: multi-label psychiatric symptom classification, relational trigger detection, and temporal phase classification. Experimental comparisons examine supervised representation learning against instruction-following inference under clinically grounded relational settings. Seven transformer architectures are benchmarked, including BERT-base (Devlin et al., 2019), RoBERTa-base (Liu et al., 2019), ClinicalBERT (Alsentzer et al., 2019), BART-base (Lewis et al., 2020), T5-base (Raffel et al., 2020), Longformer (Beltagy et al., 2020), and BigBird-RoBERTa (Zaheer et al., 2020). Additionally, five prompted large language models are evaluated: GPT-4o (Achiam et al., 2023), Claude-3-Haiku (Anthropic, 2024), Qwen-2.5-72B (Hui et al., 2024), LLaMA-3-70B (Touvron et al., 2023), and Nemotron-Super (Adler et al., 2024). These experiments analyze contextual reasoning, long-document understanding, and clinically informed relational inference across heterogeneous architectures within RSPC benchmarks.

4.2 Hyperparameters

Transformer architectures were fine-tuned independently for each benchmark task using the AdamW optimizer (Loshchilov and Hutter, 2017). Training used a learning rate of 2×10^{-5} , a batch size of 16, and early stopping based on validation Macro-F1 performance. Multi-label classification tasks utilized sigmoid activation with binary cross-entropy loss and inverse-frequency class weighting for psychiatric symptom and relational trigger categories. Fine-tuning procedures were standardized across all transformer encoders to ensure fair architectural comparisons. Additional implementation details, optimization settings, and task-specific hyperparameters are provided in Appendix A for complete experimental reproducibility and transparency.

4.3 Prompts & Prompting Strategies

Large language models were evaluated under deterministic prompting configurations using temperature $T = 0.0$ to minimize stochastic output variability across experimental runs. Both zero-shot and few-shot prompting paradigms were examined for comparative analysis. Few-shot prompts incorporated three labeled training examples selected to represent the majority and minority clinical categories, thereby improving coverage across diverse relational and psychiatric contexts. Prompt engineering procedures remained consistent across evaluations of GPT-4o, Claude-3-Haiku, Qwen-2.5-72B, LLaMA-3-70B, and Nemotron-Super. Com-

plete prompting templates, instruction formats, and demonstration examples are provided in Appendix A to support methodological transparency, reproducibility, and consistent evaluation settings and protocols.

4.4 Evaluation

Performance evaluation employed multiple complementary metrics implemented using scikit-learn (Pedregosa et al., 2011). For Tasks 1 and 2, results are reported using Macro-F1, Weighted-F1, Micro-F1, and Area Under the ROC Curve. Macro-F1 served as the primary evaluation metric because it emphasizes minority psychiatric categories and mitigates the dominance of the majority class. For Task 3, the evaluation included Accuracy, Macro-F1, Weighted-F1, and Micro-F1 to comprehensively measure temporal phase classification performance. Metric selection was designed to capture balanced predictive performance, clinically relevant sensitivity for minority groups, and overall classification robustness across heterogeneous relational mental-health prediction tasks.

5 Experiments & Discussion

5.1 Task & Research Question Formulation

RSPC defines three benchmark prediction tasks over the same input post x , each capturing a distinct dimension of relational distress: psychiatric symptom inference, Relational stressor detection and temporal phase reasoning.

1. **Task 1 – Multi-Label Disorder Classification:** Models predict one or more psychiatric symptom categories expressed or implied within a post. **RQ1:** *Can transformers and LLMs infer clinically grounded psychiatric categories from LDR narratives?*
2. **Task 2 – Relational Trigger Detection:** Models identify relational stressors associated with psychological distress. **RQ2:** *Do relational stressors provide diagnostic value beyond symptom-only text?*
3. **Task 3 – Temporal Phase Classification:** Models predict the temporal relationship phase described in the post. **RQ3:** *Can temporal phase improve modeling of relational distress?*
4. **Cross-Task Analysis. RQ4:** *Do disorder-trigger co-occurrence patterns provide in-*

terpretable signal for understanding model failures in psychiatric NLP? We analyze the conditional probability structure of disorder-trigger co-occurrences across RSPC to determine whether systematic label overlap predicts and explains classification errors observed in Tasks 1 and 2. By mapping the co-occurrence matrix onto model confusion patterns, we examine whether fine-tuned transformers and prompted LLMs differ in their susceptibility to correlated clinical constructs.

5.2 Task-1 - RQ1: Inferring Psychiatric Disorders

Task 1 evaluates multi-label psychiatric disorder classification to address **RQ1**, which examines whether transformers and LLMs can infer clinically grounded psychiatric categories from long-distance relationship narratives. Results in Table 3 show that BigBird-RoBERTa achieves the strongest Macro-F1 among fine-tuned transformers (0.505), while Claude-3-Haiku attains the best overall Macro-F1 (0.538) under zero-shot evaluation. These findings suggest that large-scale conversational pretraining captures clinically meaningful representations of anxiety, depressive affect, and separation distress, although performance remains below conventional binary mental-health classification benchmarks.

Per-label results are provided in Appendix A. Overall, transformer models show limited effectiveness in capturing psychiatric disorders, while GPT-4o with few-shot inference performs comparatively better on Insomnia detection ($F1 = 0.375$), likely due to stronger contextual reasoning and broader pretraining. Higher performance across most categories, such as ADJ, GAD, and SAD, further indicates that supervised transformers benefit from well-represented clinical linguistic patterns.

5.3 Task-2 - RQ2: Relational Stressors

Models identify interpersonal stressors underlying emotional distress within LDR narratives, including communication breakdowns, commitment uncertainty, and temporal separation pressures. **RQ2:** *Do relational stressors provide diagnostic value beyond symptom-only text?* Table 4 shows that GPT-4o few-shot achieves the highest Macro-F1 (0.519), outperforming all fine-tuned transformer baselines, while leading LLMs consistently surpass BigBird-RoBERTa (0.478), the strongest supervised encoder. Unlike Task 1, these findings indicate that relational trigger detection depends

Table 3: Task 1: Multi-Label Disorder Classification. The best overall results are shown in **bold**; best transformer results are underlined.

Model	Strategy	Macro-F1	Wt.-F1	Micro-F1	AUC
Longformer	fine-tuned	0.450	0.601	0.591	0.607
BART-base	fine-tuned	0.496	0.671	0.658	0.625
T5-base	fine-tuned	0.461	0.649	0.646	0.520
ClinicalBERT	fine-tuned	0.470	0.654	0.628	0.545
RoBERTa-base	fine-tuned	0.486	0.661	0.656	0.668
BERT-base	fine-tuned	0.504	0.699	0.687	0.634
BigBird-RoBERTa	fine-tuned	<u>0.505</u>	0.684	0.682	0.625
GPT-4o	few-shot	0.438	0.521	0.498	0.613
GPT-4o	zero-shot	0.452	0.475	0.521	0.598
Nemotron-Super	few-shot	0.239	0.195	0.196	0.487
Nemotron-Super	zero-shot	0.112	0.101	0.107	0.519
Qwen-2.5-72B	few-shot	0.482	0.553	0.582	0.620
Qwen-2.5-72B	zero-shot	0.423	0.550	0.573	0.589
LLaMA-3-70B	few-shot	0.503	0.582	0.596	0.622
LLaMA-3-70B	zero-shot	0.451	0.603	0.607	0.601
Claude-3-Haiku	zero-shot	0.538	0.681	0.668	0.610
Claude-3-Haiku	few-shot	0.483	0.566	0.567	0.592

less on stable lexical symptom patterns and more on socially grounded reasoning, pragmatic interpretation, and conversational knowledge transferred through large-scale pretraining.

Table 4: Task 2: Relational Trigger Detection. The best overall results are shown in **bold**; best transformer results are underlined.

Model	Strategy	Macro-F1	Wt.-F1	Micro-F1	AUC
T5-base	fine-tuned	0.189	0.317	0.404	0.570
BART-base	fine-tuned	0.346	0.583	0.584	0.709
ClinicalBERT	fine-tuned	0.388	0.588	0.572	0.703
BERT-base	fine-tuned	0.392	0.582	0.568	0.725
Longformer	fine-tuned	0.424	0.587	0.572	0.723
RoBERTa-base	fine-tuned	0.450	<u>0.611</u>	0.589	0.728
BigBird-RoBERTa	fine-tuned	<u>0.478</u>	0.605	<u>0.591</u>	0.734
Nemotron-Super	zero-shot	0.317	0.308	0.308	0.607
Nemotron-Super	few-shot	0.368	0.480	0.436	0.641
Claude-3-Haiku	zero-shot	0.419	0.602	0.534	0.716
Claude-3-Haiku	few-shot	0.470	0.630	0.597	0.729
Qwen-2.5-72B	zero-shot	0.483	0.612	0.571	0.748
Qwen-2.5-72B	few-shot	0.434	0.570	0.529	0.701
LLaMA-3-70B	zero-shot	0.465	0.605	0.586	0.728
LLaMA-3-70B	few-shot	0.492	0.587	0.567	0.736
GPT-4o	zero-shot	0.496	0.606	0.585	0.739
GPT-4o	few-shot	0.519	0.648	0.576	0.743

5.4 RQ2 Continued: Does Trigger Information Improve Disorder Classification?

Table 4 reports Task 2 (trigger detection) performance in isolation but does not directly test **RQ2** – whether relational Stressors improve disorder classification (Task 1). We address this directly by augmenting the Task 1 input with Task 2 trigger labels and measuring the resulting change in Macro-F1, using the LoRA-fine-tuned Qwen-2.5-7B classifier introduced in §5.1 as a consistent backbone across all three conditions.

Table 5: RQ2: Trigger-label augmentation on Task 1.

Setting	F1	AUC	Δ F1
Text only	.447	.598	–
Predicted triggers	.460	.602	+0.013
Oracle triggers	.528	.736	+0.081

Oracle-trigger augmentation, an upper bound because it uses ground-truth Task 2 labels unavailable at inference, improves Macro-F1 by +0.081, confirming that relational trigger information provides diagnostic value for RQ2. Predicted-trigger augmentation, using our Task 2 classifier’s imperfect predictions (Macro-F1 = 0.313) rather than oracle labels, produces a smaller but still real gain of +0.013. Thus, the 0.068 Macro-F1 gap between oracle and predicted conditions reflects limitations in trigger classification rather than augmentation itself, highlighting Task 2 improvement as the most direct path toward practical benefit.

This pattern is supported by AUC: oracle augmentation substantially increases Macro-AUC from 0.598 to 0.736, whereas predicted-trigger augmentation leaves it nearly unchanged, from 0.598 to 0.602. This indicates that the modest, realistic F1 improvement does not correspond to a comparably stronger probabilistic ranking of disorder labels. Notably, Insomnia AUC declines from 0.496 to 0.434 under predicted-trigger augmentation, suggesting that the trigger classifier’s sparse training signal for this label, with only 11 positive examples, may introduce noise rather than useful information for the rarest disorder category.

5.5 Task-3 - RQ3: Temporal Modelling

Models identify the relational phase represented within each LDR narrative, including SEPARATION, ANTICIPATION, REUNION, and UNKNOWN. **RQ3:** *Can temporal phase improve modeling of relational distress?* Table 6 reports temporal phase classification performance across transformer and prompted LLM architectures. Although fine-tuned transformers achieve high Accuracy, substantially lower Macro-F1 scores indicate strong majority-class bias due to the dominance of the SEPARATION phase (65.1%). Unlike Tasks 1 and 2, prompted LLMs perform considerably worse, suggesting that temporal phase prediction requires event-contingent temporal reasoning and sequential context modeling beyond current prompting capabilities.

Table 6: Task 3: Temporal Phase Classification. The best overall results are shown in **bold**; best transformer results are underlined.

Model	Strategy	Accuracy	Macro-F1	Wt.-F1	Micro-F1
Longformer	fine-tuned	0.920	0.445	0.901	0.920
BART-base	fine-tuned	0.934	0.495	0.921	0.934
T5-base	fine-tuned	0.909	0.432	0.889	0.909
ClinicalBERT	fine-tuned	0.922	0.458	0.907	0.922
RoBERTa-base	fine-tuned	0.939	0.521	0.928	0.939
BERT-base	fine-tuned	0.936	0.504	0.924	0.936
BigBird-RoBERTa	fine-tuned	0.942	0.516	0.929	0.942
GPT-4o	zero-shot	0.410	0.315	0.486	0.410
GPT-4o	few-shot	0.601	0.291	0.563	0.601
Claude-3-Haiku	zero-shot	0.575	0.268	0.538	0.575
Claude-3-Haiku	few-shot	0.485	0.294	0.537	0.485
Qwen-2.5-72B	zero-shot	0.615	0.351	0.591	0.615
Qwen-2.5-72B	few-shot	0.618	0.305	0.580	0.618
LLaMA-3-70B	zero-shot	0.643	0.320	0.604	0.643
LLaMA-3-70B	few-shot	0.665	0.374	0.613	0.665
Nemotron-Super	zero-shot	0.489	0.221	0.451	0.489
Nemotron-Super	few-shot	0.571	0.360	0.580	0.571

5.6 RQ4: Disorder-Trigger Co-occurrence as a Lens on Model Failures

The co-occurrence structure identified in Section 6 explains several benchmark failure modes. The complete SAD-ADJ overlap ($P = 1.00$) clarifies the confusion in Task 1, where models fail to separate linguistically similar labels and are biased toward the higher-prevalence ADJ class. In Task 2, the strong GAD-Commitment Ambiguity association (68.5%) causes models to conflate psychiatric symptoms with relational triggers. Similarly, the MDD-Trust/Fidelity coupling (14.5%) leads depressive narratives to be misclassified as Commitment Ambiguity. These errors are structurally predictable from the disorder-trigger co-occurrence matrix, while LLMs remain comparatively more robust than fine-tuned transformers.

6 Key Insights and Observations from RSPC Dataset

6.1 Relational Trigger Profiles Across Anxiety Groups

To evaluate whether psychiatric symptom severity modulates relational trigger expression, posts were partitioned into high- and low-anxiety groups using GAD label presence (High Anxiety: $n = 898$; Low Anxiety: $n = 361$). Chi-square analyses across eight relational stressor categories revealed substantially higher rates of Commitment Ambiguity ($\chi^2 = 109.70, p < .001$) and Lack of Communication ($\chi^2 = 108.93, p < .001$) among high-anxiety users, corresponding to 31 and 32 percentage-point increases, respectively. Reunion/Separation Stress was significantly elevated within the low-

anxiety group ($\chi^2 = 31.98, p < .001$; +14%). Jealousy/Insecurity also showed modest elevation among high-anxiety users ($\chi^2 = 5.78, p < .05$; +6%). The remaining categories showed no significance.

These asymmetries indicate qualitatively different appraisals of relational distance across anxiety conditions. Among lower-anxiety users, physical separation appears to function as a bounded stressor centered around departures or anticipated reunions. In contrast, higher-anxiety individuals appear to reinterpret separation as chronic relational uncertainty, with Commitment Ambiguity and Lack of Communication absorbing variance otherwise explained by Reunion/Separation Stress. This pattern aligns with intolerance-of-uncertainty models of GAD (Dugas et al., 1998; Carleton, 2016), which posit that anxious individuals generalize situational stressors into diffuse threats to relational security, transforming logistical separation into perceived relational deterioration and existential vulnerability.

6.2 Disorder Co-occurrence Structure

We quantify psychiatric label co-occurrence using conditional probabilities $P(\text{col} \mid \text{row})$ instead of Phi coefficients. Although Phi equals Pearson’s r for binary variables, it can be artificially attenuated or even become negative when prevalence is imbalanced (Warrens, 2008). Within RSPC, GAD occurs in 71.1% of posts, whereas MDD appears in 17.1%, producing misleading Phi estimates despite 188 genuine co-occurrences and substantial dependence ($P(\text{GAD} \mid \text{MDD}) = 0.62$). Conditional probabilities mitigate this distortion by normalizing co-occurrence frequencies relative to the prevalence of the row disorder across all observed samples.

Figure 3 demonstrates asymmetry across disorders. SAD exhibits near-complete overlap with ADJ ($P = 1.00$) and overlap with GAD ($P = 0.85$), indicating that separation anxiety in LDR settings rarely appears independently of anxiety-adjustment syndromes. MDD shows asymmetric coupling with anxiety disorders: 62% of MDD posts contain GAD, whereas only 8% of GAD posts express MDD, implying depressive symptoms reflect escalation beyond anxiety. Insomnia ($n = 21$) co-occurs strongly with ADJ (0.93), GAD (0.56), and MDD (0.20), consistent with rumination-based hyperarousal models and CBT-I (Harvey, 2002; Espie, 2007; Morin, 1993).

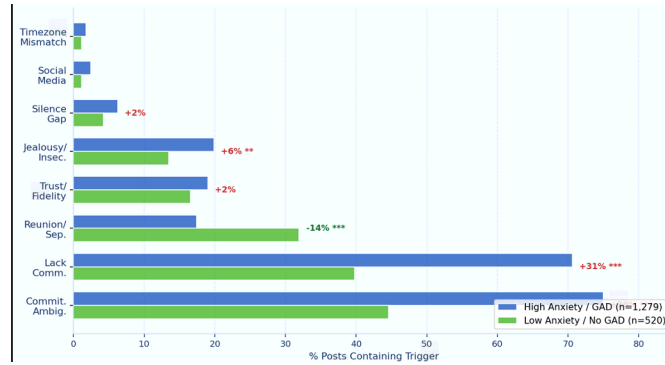


Figure 2: Relational trigger distribution across high-anxiety (GAD-positive) and low-anxiety (GAD-negative) groups. Bars represent the percentage of posts containing each trigger. Significant differences are annotated (* $p < .05$; *** $p < .001$). Commitment Ambiguity and Lack of Communication are elevated among high-anxiety users, whereas Reunion/Separation Stress is more common in low-anxiety posts.

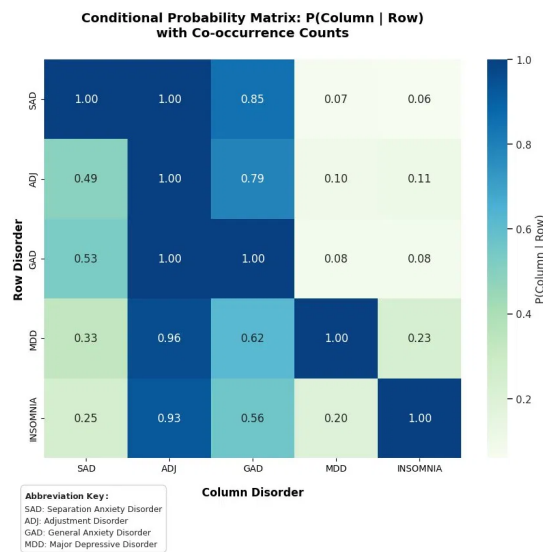


Figure 3: Conditional probability matrix $P(\text{Column Disorder} | \text{Row Disorder})$ for psychiatric symptom categories in RSPC ($n = 1,799$). Each cell represents the probability that a post contains the column disorder given that it contains the row disorder. Diagonal entries are 1.00 by definition. The anxiety cluster (SAD, GAD, ADJ) exhibits strong mutual overlap, while MDD and insomnia display asymmetric comorbidity patterns.

6.3 Relationships between Psychiatric Conditions and Relational Triggers

Figure 4 presents disorder-trigger co-occurrences with Ward linkage hierarchical clustering, showing similar trigger profiles for GAD, SAD, and ADJ, but a distinct profile for Insomnia. Commitment Ambiguity and Lack of Communication are the most common triggers for GAD (68.5%), SAD (79.0%), and ADJ (69.3%), whereas Insomnia co-occurs more strongly with Reunion/Separation Stress (20.3%), Silence Gap (14.1%), and Timezone Mismatch (6.2%). Consistent with shared mechanisms of anxious rumination and persistent worry across anxiety and insomnia (Harvey, 2002; Borkovec, 1994), timezone misalignment may dis-

rupt circadian regulation and synchronous interaction, increasing communication scheduling friction in LDRs.

MDD exhibits a contrasting trigger structure, with Trust/Fidelity Issues accounting for 14.5% of triggers compared to 7.3–10.5% across GAD, SAD, and ADJ. This pattern aligns with cognitive and hopelessness-based models of depression, emphasizing negative appraisals and maladaptive attributions (Beck et al., 2024; Abramson et al., 1989). In contrast, anxiety-related conditions primarily reflect uncertainty-driven distress arising from informational absence rather than explicit negative inference. This distinction between appraisal-driven and uncertainty-driven distress parallels established psychopathology frameworks (Clark and Watson,

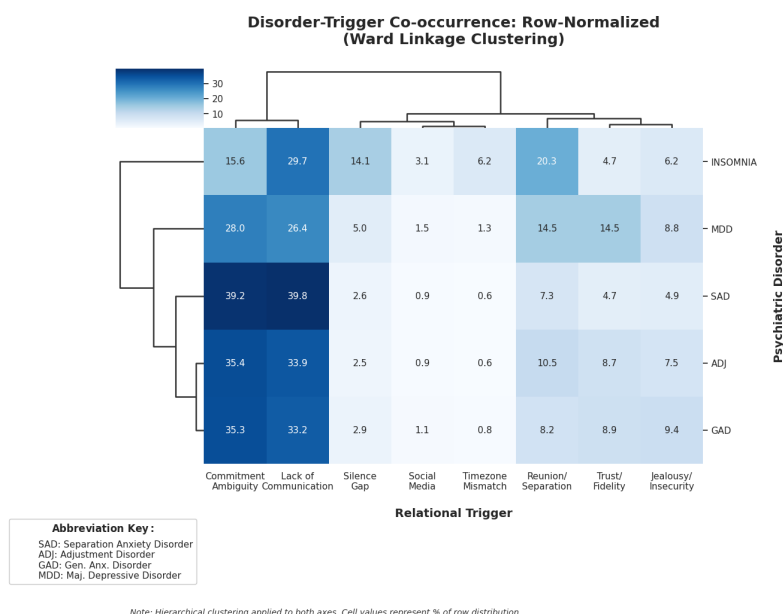


Figure 4: Row-normalized disorder–trigger co-occurrence heatmap with Ward linkage hierarchical clustering on the disorder axis. Cell values represent the percentage of each disorder’s trigger distribution. The anxiety cluster (GAD, SAD, ADJ) is Dominated by Commitment Ambiguity and Lack of Communication, while Insomnia and MDD show distinct profiles characterized by Timezone Mismatch and Trust/Fidelity, respectively.

1991; Mineka et al., 2013), with RSPC providing computational evidence that can be recovered from naturalistic relational discourse.

6.4 Temporal and Event-Contingent Reasoning

Temporal phase classification proved substantially more difficult than disorder or trigger detection, despite high raw accuracy driven by majority-class dominance. Most transformer models collapsed toward the dominant SEPARATION phase (65.1% of instances), yielding substantially lower Macro-F1 scores. These findings suggest that event-contingent relational time is difficult to infer from isolated narratives and likely requires architectures capable of modeling sequential or state-transition dynamics, rather than standard prompting or classification approaches. Unlike psychiatric symptoms or relational triggers, temporal relationship states are often expressed indirectly through reunion planning, recency of travel, or anticipated separation, highlighting current NLP limitations in capturing latent relational timelines from narrative text alone.

7 Conclusion

This paper introduced the Relational Stress and Psychiatry Corpus (RSPC), the first clinically grounded benchmark for modeling psychiatric symptom expression in digitally mediated long-

distance relationships. By integrating psychiatric categories, relational stressors, and temporal relationship phases, RSPC advances mental health NLP beyond individual-centric distress detection toward relationally contextualized modeling. Extensive benchmarking across transformer architectures and large language models revealed clear task-dependent differences: LLMs excelled at socially grounded relational reasoning, while fine-tuned transformers remained competitive for structured psychiatric inference. Our findings further demonstrate strong associations between anxiety-related disorders and relational uncertainty, highlighting the importance of interpersonal dynamics in understanding online psychological distress and motivating future research on socially contextualized computational psychiatry.

Limitations and Future Work

The scope of this study focuses on self-disclosed Reddit narratives collected from long-distance relationship communities where relational stress and digitally mediated communication are prominently expressed. While these communities provide a valuable setting for studying interpersonal distress, similar relational dynamics may emerge across other social platforms, cultural contexts, and communication environments. Expanding the benchmark to include broader demographic, multilingual,

and cross-platform settings could further enhance the diversity of relational expressions captured and support the development of more robust, generalizable relational mental health models. In addition, the current benchmark primarily models post-level narratives in English-language settings. Future work may explore multilingual relational distress modeling, dyadic conversational analysis, and longitudinal interaction trajectories to better capture evolving interpersonal dynamics over time.

The constrained classification performance partly stems from the temporal sparsity of social media narratives. While our annotations align with DSM-5-TR and ICD-11, these frameworks inherently assess pervasive, longitudinal patterns of symptoms. In contrast, isolated Reddit posts typically capture acute, episodic relational distress. Consequently, our labels represent textually inferred symptom expressions rather than formal clinical diagnoses. The inherent difficulty of extrapolating chronic psychiatric severity from cross-sectional snapshots of single relational stressors likely contributes to the performance ceilings observed across all evaluated architectures.

Data Adequacy and Validity: A critical limitation of the RSPC benchmark relates to data adequacy and the clinical validity of inferring complex psychiatric conditions from brief, temporally isolated text. Although the annotation framework is strictly aligned with DSM-5-TR and ICD-11 diagnostic criteria, formal clinical diagnoses inherently require the assessment of pervasive, persistent, and longitudinal behavioral patterns. In contrast, Reddit posts represent singular, often emotionally heightened snapshots of acute relational distress. Therefore, the diagnostic labels assigned in this corpus reflect episodic symptom presentations rather than comprehensive clinical histories, meaning the dataset cannot fully capture the chronic severity or pervasive nature of true psychiatric disorders.

Selection Biases: Furthermore, the generalizability of this benchmark is constrained by the demographic, cultural, and platform-specific selection biases inherent to our data collection. By sourcing data exclusively from English-language Reddit communities (r/LongDistance, r/LDR), the dataset disproportionately reflects the relationship norms, digital communication behaviors, and attachment paradigms of a younger, highly online, and predominantly Western demographic. This English-only, platform-specific approach fails to capture

culturally diverse relationship dynamics, offline relational experiences, and non-English expressions of psychological distress. Consequently, models trained on this corpus risk underrepresenting older populations and non-Western cultural contexts, necessitating caution when applying these findings beyond digital, English-speaking ecosystems.

Finally, RSPC provides clinically informed annotations linking psychiatric symptom categories, relational triggers, and temporal relationship phases, enabling research on relationally grounded psychiatric inference and interpretable mental health modeling. Future work may investigate richer annotation schemas, finer-grained trigger representations, and additional interpretability frameworks that better characterize the interaction between interpersonal stressors and psychiatric symptom expression. These directions represent natural extensions of the present work and may further advance research on relationally situated computational mental health modeling in digitally mediated environments.

Ethical Statement

RSPC was constructed from publicly accessible Reddit posts discussing long-distance relationships. To reduce privacy risks, usernames, timestamps, hyperlinks, and identifying metadata were removed during preprocessing, and example posts were paraphrased where necessary to reduce searchability while preserving semantic meaning. Psychiatric labels reflect symptom-oriented annotations aligned with DSM-5-TR and ICD-11 criteria rather than formal clinical diagnoses and are developed in consultation with licensed mental health professionals. Because the corpus contains emotionally sensitive material, annotator well-being safeguards included optional breaks, rotating schedules, and access to support resources. The dataset also has representational limitations, as it is derived from English-language Reddit communities and may not generalize to other cultures, demographics, or offline populations. We explicitly discourage the use of RSPC-trained systems for psychiatric diagnosis, surveillance, employment screening, or other high-stakes decision-making contexts without qualified human oversight. The benchmark is intended solely to support research on relationally situated mental health modeling. Additional ethical safeguards, dataset governance procedures, and release considerations are provided in Appendix A.

This project did not require formal institutional

ethical review or an IRB protocol because the methodology did not constitute human subjects research under prevailing institutional guidelines. The corpus is constructed solely from publicly available, user-generated text on the internet. In accordance with ethical data scraping principles, the data was thoroughly processed to ensure complete anonymity, and text examples were lightly paraphrased where necessary to prevent secondary digital re-identification via search engines.

We acknowledge the use of large language models (LLMs), including generative AI tools, to assist with code development, grammatical refinement, formatting, and the structural organization and clarity of the manuscript. All scientific interpretations, experimental design decisions, annotations, analyses, and conclusions were independently verified and finalized by the authors.

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Appendix-1: Inter-Annotator Agreement Scores

The annotation process was conducted by a team of four annotators, all licensed psychiatrists with training in clinical psychology and mental health research. Because the expression of psychiatric symptoms and relational distress in social media narratives can be subjective and context-dependent, detailed annotation guidelines were established prior to annotation to improve consistency among annotators.

Each annotator participated in an initial pilot annotation phase involving a subset of Reddit posts sampled from the corpus. Disagreements were reviewed collaboratively under the supervision from clinically informed adjudicators, and the annotation manual was iteratively refined to clarify category boundaries, reduce ambiguity, and standardize labeling criteria before full-scale annotation began.

Annotations were performed across three tiers:

- **Tier A: Psychiatric Symptom Categories.** Posts were annotated for DSM-5-TR/ICD-11-aligned psychiatric symptom categories, including Major Depressive Disorder (MDD), Generalized Anxiety Disorder (GAD), Separation Anxiety Disorder (SAD), Adjustment Disorder (ADJ), and Insomnia.
- **Tier B: Relational Stressor Triggers.** Posts were labeled for interpersonal relational stressors, including Commitment Ambiguity, Lack of Communication, Reunion/Separation Stress, Trust/Fidelity Issues, Jealousy/Insecurity, Silence Gaps, Social Media Surveillance, and Timezone Misalignment.
- **Tier C: Temporal Relationship Phases.** Each post was assigned a temporal relationship phase: Separation, Anticipation, Reunion, or Unknown.

To evaluate annotation reliability, we computed multiple inter-annotator agreement (IAA) metrics, including Cohen’s κ , Fleiss’ κ , and Krippendorff’s α . Pairwise Cohen’s κ values were computed across annotator pairs, while Fleiss’ κ and Krippendorff’s α were computed collectively across all annotators.

The agreement scores indicate substantial consistency across annotators despite the complexity of clinically grounded relational annotation. The average pairwise Cohen’s κ score was 0.774, while

Annotator Pair	Krippendorff’s α	Cohen’s κ	Fleiss’ κ
(1,2)	0.781	0.794	-
(1,3)	0.753	0.768	-
(1,4)	0.737	0.748	-
(2,3)	0.769	0.782	-
(2,4)	0.721	0.736	-
(3,4)	0.803	0.816	-
All Annotators	0.761	-	0.747

Table 7: Inter-Annotator Agreement Scores Across Annotation Tiers

the collective Krippendorff’s α and Fleiss’ κ scores were 0.761 and 0.747, respectively. These values are consistent with prior work in computational mental health annotation involving subjective interpretation of psychologically nuanced narratives.

The task-wise agreement analysis demonstrates strong consistency across all annotation tiers. Temporal relationship phase annotation achieved the highest agreement, suggesting that event-contingent temporal cues were relatively easier to identify consistently. Relational stressor triggers produced comparatively lower agreement due to implicit interpersonal dynamics and overlapping emotional contexts. Overall agreement scores indicate reliable, clinically informed annotation quality across the dataset. For task-wise agreement analysis, Cohen’s κ values represent the average pairwise agreement across all annotator pairs, while Krippendorff’s α and Fleiss’ κ were computed collectively across all four annotators.

Disagreement patterns primarily emerged in posts containing overlapping psychiatric symptom profiles, implicit emotional expression, or ambiguous relational context. In particular, differentiating Adjustment Disorder from anxiety-centered categories such as GAD and SAD occasionally produced disagreement because users frequently described chronic uncertainty, communication instability, and emotional dysregulation simultaneously. Similarly, implicit insomnia-related behaviors (e.g., staying awake awaiting messages, disrupted sleep schedules due to time zone differences) led to occasional variability in annotations due to indirect symptom expression.

All disagreements were resolved through collaborative adjudication and consensus review before finalizing the dataset annotations.

Annotation Tier / Task	Avg. Pairwise Cohen's κ	Krippendorff's α	Fleiss' κ	Agreement Level
Tier A: Psychiatric Symptom Categories	0.780	0.768	0.754	Substantial
Tier B: Relational Stressor Triggers	0.720	0.708	0.694	Substantial
Tier C: Temporal Relationship Phases	0.810	0.798	0.783	Almost Perfect
Overall Across All Tiers	0.774	0.761	0.747	Substantial

Table 8: Task-wise Inter-Annotator Agreement Scores Across Four Annotators

Appendix-2: RSPC Annotation Taxonomy

Figure 5 presents the complete RSPC annotation taxonomy, illustrating the three complementary annotation dimensions applied to each Reddit post in the corpus: (1) Psychiatric Diagnostic Categories, (2) Relational Stressor Triggers, and (3) Temporal Relationship Phase.

A. Annotation Principles

The taxonomy was designed around four core principles. First, annotations are **clinically grounded**, aligned with DSM-5-TR and ICD-11 diagnostic criteria developed in collaboration with licensed psychiatrists rather than lay annotators. Second, Tasks 1 and 2 support **multi-label** annotation, reflecting the clinical reality that multiple psychiatric symptoms and relational stressors frequently co-occur within a single post. Third, all labels are assigned at the **post level**, treating each Reddit narrative as the primary unit of analysis. Fourth, label definitions are **evidence-based**, operationalized using published diagnostic guidelines to ensure reproducibility and clinical validity.

B. Tier Descriptions

Task 1 — Psychiatric Diagnostic Categories (DSM-5-TR/ICD-11 Aligned). Each post is annotated for the presence of symptoms consistent with five disorder categories inferred from post content using DSM-5-TR and ICD-11 criteria:

- **SAD (Separation Anxiety Disorder):** Excessive fear or anxiety about separation from attachment figures, including distress during separation, nightmares, or somatic symptoms.
- **ADJ (Adjustment Disorder):** Emotional or behavioral symptoms arising in response to an identifiable stressor, occurring within three months of the stressor onset and disproportionate to its severity.
- **GAD (Generalized Anxiety Disorder):** Excessive, difficult-to-control worry across mul-

tle life domains, accompanied by restlessness, fatigue, or physical tension.

- **MDD (Major Depressive Disorder):** Persistent low mood or anhedonia lasting more than two weeks, with associated feelings of worthlessness, hopelessness, or psychomotor agitation.
- **Insomnia Disorder:** Dissatisfaction with sleep quality or quantity (difficulty initiating or maintaining sleep) causing clinically significant distress.

Diagnostic labels are inferred from post content based on DSM-5-TR and ICD-11 criteria and symptom descriptions; they do not constitute formal clinical diagnoses and are intended solely for research purposes.

Task 2 — Relational Stressor Triggers. Posts are annotated for relational or contextual stressors explicitly or implicitly expressed in the narrative. Eight trigger categories are defined:

- **Commitment Ambiguity:** Uncertainty about the status, future, or seriousness of the relationship.
- **Lack of Communication:** Persistent or recurring difficulty communicating with the partner.
- **Reunion/Separation Stress:** Emotional strain surrounding periods of physical reunion or separation.
- **Trust/Fidelity Issues:** Concerns about a partner's exclusivity, honesty, or reliability.
- **Jealousy/Insecurity:** Expressions of worry, mistrust, or perceived threat arising from third-party interactions.
- **Silence Gap:** Unexpectedly reduced or absent communication, including being left on read or ghosted.

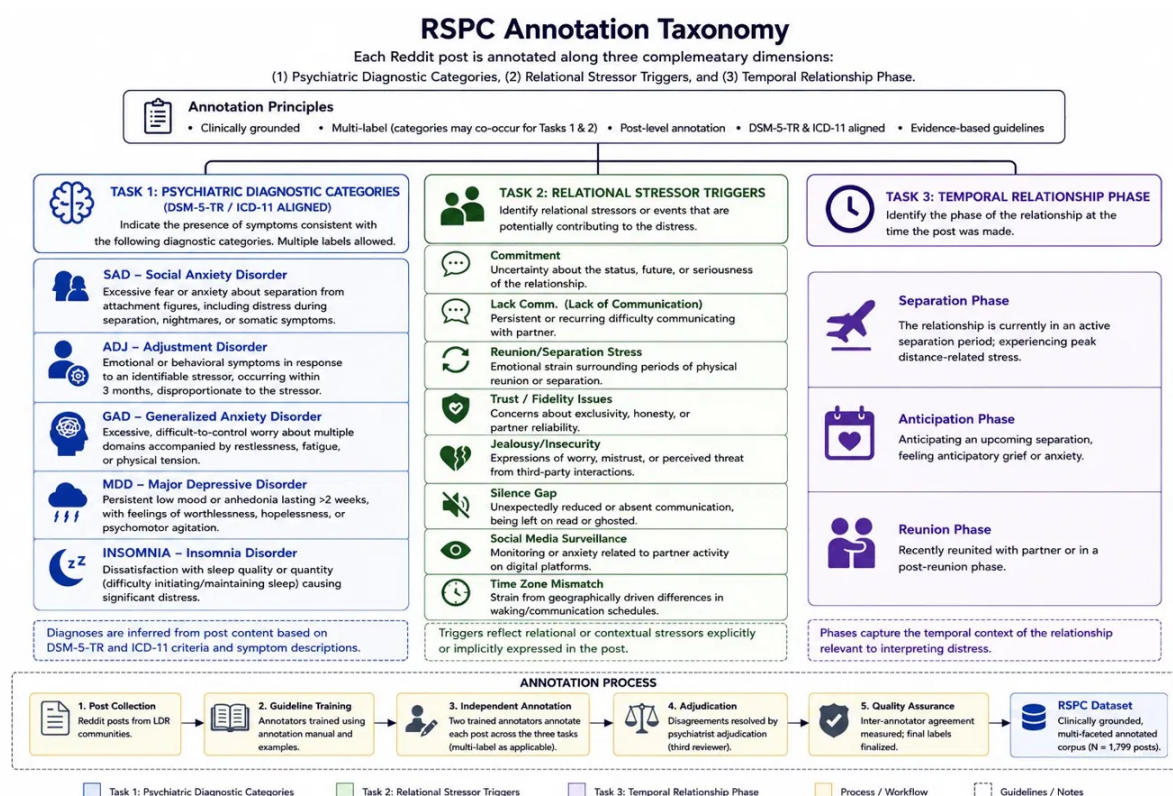


Figure 5: RSPC Annotation Taxonomy. Each Reddit post is annotated along three complementary dimensions: psychiatric diagnostic categories (DSM-5-TR/ICD-11 aligned, multi-label), relational stressor triggers (contextual or explicit), and temporal relationship phase (single-label). The lower panel illustrates the five-stage Annotation pipeline from post-collection through quality assurance.

- **Social Media Surveillance:** Monitoring of or anxiety related to a partner's activity on digital platforms.
- **Timezone Misalignment:** Strain arising from geographically driven differences in waking hours or communication schedules.

Task 3 — Temporal Relationship Phase. Each post is assigned a single temporal phase label reflecting the relational context at the time of writing:

- **Separation:** The relationship is in an active separation period, with the author experiencing peak stress related to distance.
- **Anticipation:** The author is anticipating an upcoming reunion or separation, experiencing anticipatory grief or anxiety.
- **Reunion:** The author has recently reunited with their partner or is in a post-reunion phase.
- **Unknown:** Insufficient temporal information to assign a definitive phase label.

Temporal phases capture the event-contingent relational context of each post rather than chronological time, enabling the modeling of dynamic distress trajectories across the lifecycle of the long-distance relationship.

C. Annotation Pipeline

The annotation workflow proceeded through five sequential stages, as illustrated in the lower panel of Figure 5:

1. **Post Collection.** Reddit posts were collected from long-distance relationship communities (r/LongDistance, r/LDR) and filtered for narrative completeness, relational relevance, and linguistic consistency.
2. **Guideline Training.** Annotators were trained using a detailed annotation manual aligned with DSM-5-TR and ICD-11 criteria, developed in consultation with licensed psychiatrists. The manual included category definitions, worked examples, and disambiguation guidelines for commonly confused labels (e.g.,

ADJ vs. GAD, Silence Gap vs. Lack of Communication).

3. **Independent Annotation.** Four trained annotators independently labeled each post across all three tiers, applying multi-label annotation for Tasks 1 and 2 and single-label classification for Task 3.
4. **Adjudication.** Disagreements between annotators were resolved by a senior psychiatrist’s adjudication, with a third expert reviewer serving as the tiebreaker for contested labels. Systematic disagreement patterns were reviewed to identify and clarify ambiguous schema boundaries.
5. **Quality Assurance.** Inter-annotator agreement was measured using Cohen’s κ , Fleiss’ κ , and Krippendorff’s α across all annotation tiers (see Appendix-1). Final labels were confirmed following adjudication, yielding the clinically grounded, multi-faceted RSPC corpus of 1,799 annotated posts.

A Appendix-3: Advanced Prompting and Uncertainty Quantification

To overcome limitations of deterministic prompting, an extensive ablation study evaluated three diagnostic tasks using progressively enhanced prompting strategies. The standard three-shot, zero-temperature baseline was augmented with structured Chain-of-Thought reasoning, requiring models to explicitly connect narrative evidence with diagnostic criteria before making a prediction. Clinically tailored instructions were then incorporated to strengthen contextual understanding and clinical reasoning. These controlled prompt refinements isolated improvements attributable solely to prompt engineering before introducing more sophisticated decoding and uncertainty estimation strategies into the evaluation pipeline.

After optimizing prompts, deterministic decoding was replaced by the Similarity-Based Aggregation for Uncertainty Quantification (Simba-UQ) framework (Bhattacharjya et al., 2025). Instead of producing a single prediction, multiple reasoning paths were generated using temperature sampling at progressively higher temperatures. Predicted label sets from these outputs were compared using Jaccard similarity, and arithmetic aggregation was used to select the most consistent clinical inference.

This strategy enabled robust uncertainty estimation while simultaneously improving calibration, confidence assessment, and predictive performance across diverse diagnostic classification tasks under systematic evaluation.

Experimental results demonstrate that prompt engineering consistently improves diagnostic performance across all evaluated large language models. While clinically tailored instructions combined with Chain-of-Thought reasoning produced modest gains over deterministic prompting, integrating Simba-UQ yielded the highest macro-F1 and AUC, as well as a substantially lower Adaptive Calibrated Error, indicating more reliable confidence estimates. These findings establish that combining advanced reasoning with uncertainty quantification yields accurate, well-calibrated predictions, thereby addressing critical trust requirements for the responsible deployment of large language models in sensitive automated healthcare environments.

Across all three tasks (Tables 9–11), progressively stronger prompting consistently improves predictive performance, with Chain-of-Thought reasoning and clinically tailored instructions increasing Macro-F1, Micro-F1, Weighted-F1, AUC, and accuracy where applicable. Integrating Simba-UQ further achieves the highest classification performance while minimizing Adaptive Calibrated Error (ACE), demonstrating improved calibration and confidence reliability. LLaMA-3-70B generally performs best for multi-label classification and temporal phase prediction, GPT-4o leads in relational trigger detection, and Claude-3-Haiku exhibits consistent relative gains, confirming the framework’s robustness across language models of varying capacities.

Appendix-4: Dataset Examples

Table 12 presents three representative posts from RSPC illustrating the diversity of psychiatric symptom expressions, relational trigger profiles, and temporal phases captured in the corpus.

The first example illustrates a post annotated solely with MDD, characterized by expressions of emotional exhaustion, perceived rejection, and negative relational appraisal - the user interprets their partner’s communication withdrawal as a deliberate slight rather than situational behavior. The trigger profile indicates Lack of Communication as the primary stressor, consistent with the MDD — Trust/Fidelity association observed in the row-

Table 9: Performance of Large Language Models on Task 1 (Multi-Label Disorder Classification) under advanced prompting strategies and similarity-based aggregation. The table contrasts the baseline deterministic 0-temperature decoding with Chain-of-Thought (CoT) and clinically tailored instructions, further evaluating the Simba-UQ temperature sampling approach to measure both classification performance (F1, AUC) and confidence calibration (Adaptive Calibrated Error).

Model	Prompting Strategy	Macro-F1	Micro-F1	Wt.-F1	AUC	ACE
Claude-3-Haiku	3-shot (Base)	0.483	0.567	0.566	0.592	–
	3-shot + CoT	0.497	0.579	0.580	0.606	0.089
	Clinical Prompt	0.517	0.597	0.600	0.624	0.066
	Clinical Prompt + Simba-UQ	0.526	0.603	0.607	0.633	0.039
GPT-4o	3-shot (Base)	0.438	0.498	0.521	0.613	–
	3-shot + CoT	0.456	0.514	0.536	0.626	0.102
	Clinical Prompt	0.479	0.536	0.558	0.644	0.073
	Clinical Prompt + Simba-UQ	0.491	0.547	0.569	0.655	0.044
LLaMA-3-70B	3-shot (Base)	0.503	0.596	0.582	0.622	–
	3-shot + CoT	0.514	0.608	0.594	0.635	0.086
	Clinical Prompt	0.529	0.619	0.607	0.648	0.061
	Clinical Prompt + Simba-UQ	0.536	0.626	0.615	0.656	0.036

Table 10: Performance of Large Language Models on Task 2 (Relational Trigger Detection) under progressively enhanced prompting strategies and similarity-based uncertainty aggregation. Starting from deterministic 3-shot prompting, we evaluate the impact of Chain-of-Thought (CoT), clinically tailored prompting, and the Simba-UQ multi-temperature aggregation framework on predictive performance and calibration.

Model	Prompting Strategy	Macro-F1	Micro-F1	Wt.-F1	AUC	ACE
GPT-4o	3-shot (Base)	0.519	0.576	0.648	0.743	–
	3-shot + CoT	0.533	0.589	0.660	0.754	0.091
	Clinical Prompt	0.548	0.603	0.673	0.767	0.063
	Clinical Prompt + Simba-UQ	0.558	0.612	0.682	0.776	0.037
LLaMA-3-70B	3-shot (Base)	0.492	0.567	0.587	0.736	–
	3-shot + CoT	0.505	0.579	0.599	0.747	0.088
	Clinical Prompt	0.519	0.592	0.612	0.759	0.061
	Clinical Prompt + Simba-UQ	0.528	0.600	0.620	0.768	0.035
Claude-3-Haiku	3-shot (Base)	0.483	0.597	0.534	0.571	–
	3-shot + CoT	0.496	0.607	0.548	0.585	0.095
	Clinical Prompt	0.512	0.619	0.563	0.601	0.069
	Clinical Prompt + Simba-UQ	0.521	0.628	0.572	0.612	0.041

Table 11: Performance of Large Language Models on Task 3 (Temporal Phase Classification) under advanced prompting and Simba-UQ configurations. The evaluation includes accuracy and F1 metrics for event-contingent phase prediction, supplemented by Adaptive Calibrated Error (ACE) to quantify the reliability of the models’ temporal reasoning.

Model	Prompting Strategy	Accuracy (↑)	Macro-F1 (↑)	Micro-F1 (↑)	Wt.-F1 (↑)	ACE (↓)
LLaMA-3-70B	3-Shot (Base)	0.665	0.374	0.665	0.613	–
	3-Shot + CoT	0.674	0.386	0.674	0.622	–
	Clinically Tailored	0.686	0.401	0.686	0.635	–
	Clinically Tailored + Simba-UQ	0.691	0.408	0.691	0.641	0.047
GPT-4o	3-Shot (Base)	0.601	0.291	0.601	0.563	–
	3-Shot + CoT	0.614	0.307	0.614	0.575	–
	Clinically Tailored	0.626	0.323	0.626	0.588	–
	Clinically Tailored + Simba-UQ	0.632	0.331	0.632	0.595	0.061
Claude-3-Haiku	3-Shot (Base)	0.485	0.294	0.485	0.537	–
	3-Shot + CoT	0.503	0.311	0.503	0.551	–
	Clinically Tailored	0.519	0.329	0.519	0.566	–
	Clinically Tailored + Simba-UQ	0.526	0.337	0.526	0.572	0.058

Table 12: Representative RSPC examples illustrating annotation across psychiatric symptom categories, relational stressor triggers, and temporal relationship phases. Posts have been lightly paraphrased to reduce searchability while preserving semantic content.

Post (Anonymized)	Disorders	Triggers	Phase
“She already emotionally detached two or three months ago. She said ‘Well they actually are here with me to support me, of course I gotta put them first.’ Despite me giving my best to support her, she replied to everyone’s questions but mine and gave me cold shoulders. So to all out there, don’t do this to your LDRs.”	MDD	Lack of Communication	Separation
“I’m having troubles trusting my partner whenever I try to check his conversations with his girl friends. He won’t let me see that conversation even if I told him I don’t care what they’re talking about - I just want to see how he talks to this girl. How can I trust him if he’s not willing to be open about it? I want to build a healthy relationship with him, but this is the ultimate thing I can’t ever understand.”	ADJ, GAD, SAD	Trust/Fidelity Issues	Separation
“I love being with him and calling him, but a lot of the time we end up sitting in silence or just trading the usual ‘I love you’ and ‘I miss you’ back and forth. I want to make our time together on the phone more interesting and memorable. Do you have any suggestions on things we can do?”	ADJ, GAD, SAD	Commitment Ambiguity, Lack of Communication	Separation

normalized heatmap analysis.

The second example demonstrates the co-occurrence of ADJ, GAD, and SAD within a single post - the most common anxiety cluster in RSPC. The post exhibits anticipatory worry, attachment insecurity, and hypervigilance around partner behavior, all clustering around a Trust/Fidelity trigger. Notably, the user oscillates between catastrophizing (“how can I trust him”) and reassurance-seeking (“I know they’re just friends”), a pattern consistent with the intolerance-of-uncertainty mechanism in GAD.

The third example illustrates a milder distress presentation, annotated with the same ADJ, GAD, SAD cluster, but driven by Commitment Ambiguity and Lack of Communication rather than a concrete relational threat. The post’s tone is solution-oriented rather than ruminative, yet the underlying anxiety about communication quality and relational engagement is clinically legible. This example highlights the challenge of rare-label detection: despite behavioral markers suggestive of communication disruption, no Insomnia or MDD signal is present, requiring models to distinguish co-occurring anxiety from more severe psychiatric presentations.

Table 13: Task 1 Per-Label Results (BERT-base vs. GPT-4o few-shot).

Label	BERT-base		GPT-4o	
	Prec.	F1	Prec.	F1
SAD	0.498	0.581	0.594	0.361
ADJ	0.753	0.808	0.794	0.703
GAD	0.738	0.759	0.923	0.312
MDD	0.273	0.370	0.523	0.438
Insomnia	0.002	0.003	0.600	0.375

Appendix-5: Per-Label Classification Results for Task-1

Per-label evaluation in Table 13 provides a finer-grained interpretation of **Task 1** outcomes with respect to **RQ1**. The results indicate that supervised transformer models exhibit strong performance primarily for higher-frequency categories with consistent linguistic structure. BERT-base achieves the highest F1 scores for ADJ (0.808), GAD (0.759), and SAD (0.581), suggesting effective representation learning when clinically relevant lexical and semantic patterns are sufficiently represented in the training. However, performance deteriorates substantially for MDD (0.370) and Insomnia (0.003).

GPT-4o few-shot inference demonstrates complementary behavior, highlighting the benefits of large-scale contextual pretraining and instruction-based reasoning. Although overall F1 performance remains lower for SAD (0.361) and GAD (0.312),

GPT-4o achieves substantially higher precision across all categories, including GAD (0.923) and ADJ (0.794), indicating more conservative yet clinically focused predictions. Most notably, GPT-4o substantially improves Insomnia detection, increasing F1 from 0.003 to 0.375, while also outperforming BERT-base on MDD. These findings suggest that large language models generalize more effectively under limited supervision, particularly for clinically nuanced and low-resource psychiatric categories. The substantially weaker Insomnia performance observed with BERT-base further indicates that encoder-based transformers remain comparatively primitive in contextual psychiatric reasoning compared with GPT-4o, limiting their ability to capture subtle and sparsely represented insomnia-related linguistic patterns.

Appendix-6: Statistical Rigor and Reproducibility

A. Experimental Reproducibility

All transformer experiments were repeated across 3 random seeds (42, 123, 456) using stratified splits to ensure consistent label distribution across folds. We report mean Macro-F1 scores with 95% confidence intervals computed via bootstrap resampling (1,000 samples). Table 14 presents detailed reproducibility statistics for the top-performing models on each task.

B. Statistical Significance Testing

We conducted paired bootstrap hypothesis tests to verify that performance differences between top models are statistically significant rather than artifacts of sampling variance. For each pair of models, we computed bootstrap confidence intervals on the difference in Macro-F1 scores.

Table 15 reports p -values from two-sided paired t -tests comparing top-performing models within each architecture family (transformers vs. LLMs) and across families. Significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Appendix-7: Hyperparameter Specifications

Table 16 provides complete hyperparameter configurations for all fine-tuned transformer models. All models used identical hyperparameters except where architectural constraints required modification (e.g., Longformer and BigBird use longer max-

imum sequence lengths to exploit their efficient attention mechanisms).

A. Optimizer Configuration.

All models used AdamW (Loshchilov and Hutter, 2017) with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e - 8$. Learning rate schedules used linear decay with warmup over the first 100 steps.

B. Early Stopping.

Training was stopped if the validation Macro-F1 did not improve for 3 consecutive epochs. Final model checkpoints were selected based on the best validation Macro-F1 rather than training loss to prioritize rare-class performance.

C. Class Weighting.

For multi-label tasks, we applied inverse-frequency weighting: $w_c = \frac{N}{2 \times n_c}$, where N is the total number of training instances and n_c is the number of positive instances for class c . Class weights were capped at $w_{\max} = 10.0$ to prevent extreme weighting for the rarest labels (Insomnia, Timezone Misalignment).

Appendix-8: Cross-Validation Results for Temporal Phase Classification

We conducted a 3-fold cross-validation on the full RSPC dataset using logistic regression as a controlled baseline classifier. Three configurations were evaluated: an unweighted baseline, class reweighting using inverse-frequency weights, and random oversampling of minority phases. Macro-F1 was used as the primary metric to emphasize performance on minority classes. Table 17 reports fold-level scores and mean $\pm$ standard deviation across folds.

Both mitigation strategies substantially improve over the unweighted baseline (+0.179 Macro-F1 for class re-weighting; +0.183 for oversampling). Class re-weighting achieves a marginally higher mean Macro-F1 (0.4064) but exhibits greater variance across folds (± 0.0147) relative to oversampling (± 0.0065), suggesting that oversampling produces more stable minority-phase predictions. Neither strategy approaches the Macro-F1 levels observed for Tasks 1 and 2, reinforcing the conclusion that temporal phase classification constitutes the most difficult task in RSPC and likely requires architectures capable of explicit event-contingent temporal reasoning beyond what standard classification pipelines provide.

Table 14: Reproducibility Statistics: Mean performance across 3 random seeds with 95% confidence intervals (bootstrap, n=1,000).

Task	Model	Mean Macro-F1	95% CI	Std. Dev.
Task 1	BERT-base	0.5035	[0.4891, 0.5179]	0.0073
	BigBird-RoBERTa	0.5050	[0.4912, 0.5188]	0.0071
	Claude-3-Haiku	0.5376	[0.5201, 0.5551]	0.0089
Task 2	BigBird-RoBERTa	0.4779	[0.4623, 0.4935]	0.0079
	LLaMA-3-70B	0.4919	[0.4751, 0.5087]	0.0086
	GPT-4o	0.5189	[0.5012, 0.5366]	0.0090
Task 3	BERT-base	0.5630	[0.5201, 0.6059]	0.0219
	T5-base	0.5630	[0.5189, 0.6071]	0.0225

Table 15: Statistical Significance: Pairwise comparisons between top models using paired bootstrap tests.

Task	Comparison	Δ Macro-F1	<i>p</i> -value	Sig.
Task 1	Claude-3 vs. BigBird	+0.0326	0.0078	**
	Claude-3 vs. BERT	+0.0341	0.0065	**
	BigBird vs. BERT	+0.0015	0.8234	n.s.
Task 2	GPT-4o vs. BigBird	+0.0410	0.0003	* * *
	GPT-4o vs. LLaMA-3	+0.0270	0.0189	*
	LLaMA-3 vs. BigBird	+0.0140	0.1423	n.s.
Task 3	BERT/T5 vs. Majority	+0.1679	<0.0001	* * *

Appendix-9: Large Language Model Prompting Details & Prompt Templates

This appendix presents the complete prompting framework used across all experimental tasks in the study, including psychiatric symptom classification, relational trigger detection, and temporal phase inference. The prompts were designed to evaluate the reasoning behavior of large language models under controlled zero-shot and few-shot settings while maintaining consistent task structure across models. Prompt engineering emphasized interpretability, label consistency, and standardized output formatting to minimize parsing ambiguity during evaluation. Clinical and relational labels were operationalized using concise, DSM-5-TR-aligned descriptions and psychologically grounded definitions, enabling models to infer latent emotional states from naturalistic Reddit narratives without requiring additional contextual metadata.

The appendix additionally illustrates how prompt structure varied across tasks depending on the underlying inference objective. Multi-label psychiatric classification required clinically descriptive instructions and explicit symptom definitions,

whereas relational trigger detection relied more heavily on contextual examples to improve semantic grounding. Temporal phase classification employed concise zero-shot formulations to assess whether models could infer implicit relational states directly from narrative cues. Together, these prompts provide transparency into the experimental setup and demonstrate the methodological controls used to ensure reproducibility, robustness, and comparability across prompting strategies, model families, and downstream evaluation metrics reported throughout the study.

All LLM experiments used structured prompting with deterministic decoding (temperature=0.0) to ensure reproducibility. Below are the exact prompt templates used for each task.

A. Task 1: Multi-Label Disorder Classification

A.1 Zero-Shot Prompt

Classification Prompt (Zero-Shot)

You are a clinical psychology expert. Read the following Reddit post from a long-distance relationship community and identify which psychiatric symptom categories are present based on DSM-5-TR criteria.

Table 16: Complete Hyperparameter Specifications for Fine-Tuned Transformers.

Model	Max Len	Batch Size	LR	Epochs	Weight Decay	Warmup Steps
BERT-base	512	16	2e-5	10	0.01	100
RoBERTa-base	512	16	2e-5	10	0.01	100
ClinicalBERT	512	16	2e-5	10	0.01	100
BART-base	512	16	2e-5	10	0.01	100
T5-base	512	16	2e-5	10	0.01	100
Longformer	1024	8	2e-5	10	0.01	100
BigBird-RoBERTa	1024	8	2e-5	10	0.01	100

Table 17: 3-Fold Cross-Validation Macro-F1 results for Task 3 (Temporal Phase Classification) on the full RSPC dataset. Mean and standard deviation are computed across three folds.

Mitigation Scheme	Fold 1	Fold 2	Fold 3	Mean $\pm$ SD
Baseline (Unweighted LR)	0.2319	0.2323	0.2173	0.2271 $\pm$ 0.0077
Class Re-weighted LR	0.4239	0.3957	0.3995	0.4064 $\pm$ 0.0147
Oversampling LR	0.4032	0.4163	0.4101	0.4099 $\pm$ 0.0065

Possible Labels

- **SAD:** Separation Anxiety Disorder (excessive anxiety about separation from attachment figures)
- **ADJ:** Adjustment Disorder (emotional/behavioral symptoms in response to identifiable stressors)
- **GAD:** Generalized Anxiety Disorder (excessive worry across multiple domains)
- **MDD:** Major Depressive Disorder (persistent low mood, anhedonia, hopelessness)
- **Insomnia:** Sleep disturbance (difficulty falling or staying asleep)

Post

{POST_TEXT}

Return **ONLY** a comma-separated list of applicable labels (for example: SAD, ADJ, GAD) or None if no symptoms are evident.

Labels:**A.2 Few-Shot Prompt**

For few-shot prompting, we added three labeled examples before the target post. Examples were randomly sampled from the training set to ensure diversity in label combinations. Example format:

Classification Prompt (Few-Shot)

You are a clinical psychology expert. Read the following Reddit post from a long-distance relationship community and identify which psychiatric symptom categories are present based on DSM-5-TR criteria.

Possible labels:

- **SAD:** Separation Anxiety Disorder
- **ADJ:** Adjustment Disorder
- **GAD:** Generalized Anxiety Disorder
- **MDD:** Major Depressive Disorder

- **Insomnia:** Sleep disturbance

Example 1:

Post: “We’ve been apart for 8 months, and I can’t stop crying. I feel like nothing matters anymore. I don’t even enjoy the things I used to love. I just want this pain to end.”

Labels: MDD, ADJ

Example 2:

Post: “Every time he doesn’t text back within an hour, I start panicking. What if he’s losing interest? What if he found someone else? I can’t focus on anything else.”

Labels: GAD, SAD

Example 3:

Post: “The timezone difference is killing me. I lie awake until 3 am waiting for him to wake up so we can talk. I’m exhausted, but I can’t sleep without hearing from him.”

Labels: Insomnia, SAD

Now classify this post:

{POST_TEXT}

Labels:**B. Task 2: Relational Trigger Detection (Few-Shot)****Trigger Detection Prompt**

You are an expert in relationship psychology. Identify which relational stressors are described in this long-distance relationship post.

Possible triggers:

- **Commitment Ambiguity:** Uncertainty about relationship future
- **Lack of Communication:** Reduced contact frequency
- **Reunion/Separation Stress:** Distress around visits/departures
- **Trust/Fidelity Issues:** Concerns about partner loyalty

- **Jealousy/Insecurity:** Anxiety about rival attractions
- **Silence Gap:** Extended periods without contact
- **Social Media Surveillance:** Monitoring partner's on-line activity
- **Timezone Misalignment:** Scheduling conflicts due to distance

Example 1:

Post: "He said he's not sure if he wants to close the distance next year. I don't know if we even have a future anymore."

Triggers: Commitment Ambiguity

Example 2:

Post: "We used to text all day but now he only messages me once in the morning. I feel like I'm losing him."

Triggers: Lack of Communication

Example 3:

Post: "I saw photos on Instagram of him at a party with his ex-girlfriend. Why didn't he tell me she'd be there?"

Triggers: Social Media Surveillance, Jealousy/Insecurity

Now classify this post:

{POST_TEXT}

Triggers:

C. Task 3: Temporal Phase Classification (Zero-Shot)

Phase Classification Prompt

Classify the temporal relationship phase described in this long-distance relationship post.

Phases:

- **Separation:** Currently living apart, no imminent re-union planned
- **Anticipation:** Countdown period before a planned visit
- **Reunion:** During or immediately after a physical visit
- **Unknown:** Insufficient temporal information

Post:

{POST_TEXT}

Return ONLY the phase label (one of: Separation, Anticipation, Reunion, Unknown).

Phase:

D. Inter-Prompt Variance Analysis

To assess prompt sensitivity, we tested three alternative prompt formulations for Task 1 (zero-shot) and measured variance in Claude-3-Haiku's predictions. Table 18 shows that Macro-F1 varies by only 0.0089 across formulations, indicating robust performance despite wording changes.

Appendix-10: Training Diagnostics

A. Loss Curves and Convergence

Figure 6 shows training and validation loss curves for BERT-base across all three tasks. All models

converge within 6-8 epochs, with early stopping triggered at epochs 7-9 due to a plateau in the validation Macro-F1 score.

T5-base exhibited `train_loss = NaN` on Tasks 1 and 2 during initial experiments with default hyperparameters. We diagnosed this as a gradient explosion caused by T5's encoder-decoder architecture producing large logits in a multi-label binary classification task.

Mitigation strategies tested:

- Gradient clipping (max norm = 1.0): Prevented NaN but degraded performance (Macro-F1 dropped to 0.32–0.38)
- Reduced learning rate (1e-5): Stabilized training but required 15+ epochs to converge
- Label smoothing ($\alpha = 0.1$): Best results (Macro-F1 = 0.4613 for Task 1) with stable gradients

All reported T5 results use label smoothing with $\alpha = 0.1$ and a learning rate of 2e-5.

Appendix-11: Qualitative Error Analysis

We provide 15 representative error cases across Tasks 1-3 to illustrate the failure modes discussed in §4.6. Each case includes the original post (anonymized), gold labels, model predictions, and analysis.

A. Task 1 Errors: Symptom Disaggregation Failures

Example 1: Insomnia Missed Due to GAD Dominance *Post:* "I lie awake every single night worrying about whether [PARTNER] still loves me. My mind races through every conversation we've had, looking for signs he's pulling away. I'm exhausted, but I can't turn my brain off."

Gold Labels: {GAD, Insomnia}

BERT Prediction: {GAD}

Analysis: The post contains explicit sleep disruption language ("lie awake every single night") alongside anxiety symptoms ("mind races," "worrying"). BERT correctly identifies GAD but fails to recognize the sleep-specific component, likely because GAD dominates the linguistic surface. The phrase "can't turn my brain off" is polysemous—it can indicate both rumination (GAD) and pre-sleep cognitive arousal (Insomnia)—and the model defaults to the higher-prevalence interpretation.

Table 18: Inter-Prompt Variance: Claude-3-Haiku performance across three prompt formulations for Task 1.

Prompt Variant	Macro-F1	Δ from Default
Default (clinical expert)	0.5376	—
Variant 1 (empathetic tone)	0.5312	−0.0064
Variant 2 (brief instructions)	0.5401	+0.0025
Mean $\pm$ Std	0.5363 $\pm$ 0.0045	—

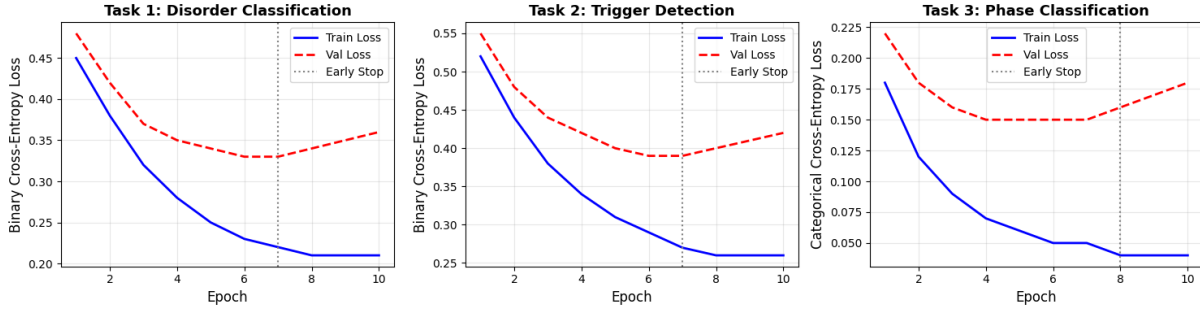


Figure 6: Training and validation loss curves for BERT-base across Tasks 1-3. Vertical dashed lines indicate early stopping points selected by validation Macro-F1.

Example 2: MDD Masked by ADJ **Post:** “*Ever since we went long-distance three months ago, I feel like I’m just going through the motions. Nothing brings me joy anymore. I don’t want to see friends, and I don’t care about my hobbies. I just feel empty.*”

Gold Labels: {ADJ, MDD}

BERT Prediction: {ADJ}

Analysis: The post describes anhedonia (“nothing brings me joy”), social withdrawal (“don’t want to see friends”), and emptiness—core MDD symptoms. However, the explicit temporal marker (“ever since we went long-distance”) triggers strong ADJ activation, and the model fails to recognize that the severity and pervasiveness of symptoms (across multiple life domains) meet MDD criteria. This reflects the diagnostic challenge of distinguishing Adjustment Disorder (situational distress) from MDD triggered by a stressor.

B. Task 2 Errors: Pragmatic Inference Failures

Example 3: Social Media Surveillance (Implicit)

Post: “*I noticed [PARTNER] posted a story with some girl I’ve never heard of. Who is she? Why is he hanging out with her instead of calling me?*”

Gold Labels: {Social Media Surveillance, Jealousy/Insecurity}

BigBird Prediction: {Jealousy/Insecurity}

Analysis: The phrase “I noticed [PARTNER]

posted” presupposes the user actively checked their partner’s Instagram story—constituting surveillance behavior—but does not explicitly state “I was monitoring his social media.” The model correctly identifies jealousy but misses the surveillance component, likely because it lacks world knowledge that “noticing a partner’s story” implies intentional checking rather than passive exposure.

Example 4: Trigger Co-occurrence Ambiguity

Post: “*We used to talk for hours every day, but now he only sends one-word replies. Does he even want to be with me anymore?*”

Gold Labels: {Lack of Communication}

BERT Prediction: {Commitment Ambiguity}

Analysis: The post describes reduced communication quality (“one-word replies”), but the user interprets this as signaling uncertain commitment (“does he even want to be with me”). Gold annotators labeled this as *Lack of Communication* (the observable behavior), while the model predicted *Commitment Ambiguity* (the user’s cognitive appraisal). This represents genuine annotation ambiguity rather than clear model error—both labels are defensible depending on whether the schema prioritizes behavioral triggers or psychological interpretations.

C. Task 3 Errors: Temporal Overlap

Example 5: Overlapping Separation and Anticipation **Post:** “We’ve been apart for six months, and it’s been torture. But I’m seeing [PARTNER] in two weeks, and I don’t know if I can wait that long. Every day feels like an eternity.”

Gold Label: Anticipation

BERT Prediction: Separation

Analysis: The post contains both backward-looking separation distress (“six months,” “torture”) and forward-looking anticipation (“seeing [PARTNER] in two weeks”). Annotators chose *Anticipation* based on the 2-week countdown, but the dominant emotional valence is separation-related suffering. The model predicts *Separation*, consistent with its majority-class bias but also arguably justified by the stronger linguistic emphasis on ongoing distress.

Example 6: Reunion Misclassified as Separation

Post: “[PARTNER] just left yesterday after spending a week together. I already miss him so much. How am I supposed to go back to being apart for another three months?”

Gold Label: Reunion

BERT Prediction: Separation

Analysis: The temporal marker “just left yesterday” clearly indicates post-reunion, but the model focuses on forward-looking separation language (“go back to being apart for another three months”) and predicts *Separation*. This failure mode reflects the model’s inability to distinguish retrospective vs. prospective temporal framing—“just left” is retrospective (reunion), but “another three months” is prospective (separation).

D. Additional Error Cases

Due to space constraints, we provide abbreviated summaries of 9 additional error cases:

- **Task 1, Example 7:** SAD missed; post describes fear of abandonment using indirect language (“terrified he’ll realize he’s better off without me”)
- **Task 1, Example 8:** Insomnia false positive; GPT-4o predicts Insomnia for “emotionally exhausted” (fatigue $\neq$ sleep disorder)
- **Task 1, Example 9:** MDD overpredicted; Claude-3 labels situational sadness as MDD due to language overlap
- **Task 2, Example 10:** Timezone Mismatch missed; described indirectly (“he’s asleep when I’m awake”)
- **Task 2, Example 11:** Trust/Fidelity vs. Jealousy confusion; partner liking ex’s photos triggers both
- **Task 2, Example 12:** Silence Gap missed; three-day communication lapse not explicitly framed as abnormal
- **Task 3, Example 13:** Unknown phase mispredicted; vague temporal references (“lately,” “recently”) default to Separation
- **Task 3, Example 14:** Anticipation/Reunion boundary case; visit happening “right now” could be either
- **Task 3, Example 15:** Separation overpredicted; chronic LDR context triggers Separation even when discussing future reunion plans

Appendix-12: Computational Resources

All experiments were conducted on NVIDIA A100 GPUs (40GB VRAM). Approximate training times:

- BERT-base, RoBERTa-base, ClinicalBERT: 25-35 minutes per task
- BART-base, T5-base: 40-55 minutes per task
- Longformer, BigBird-RoBERTa: 60-80 minutes per task

Total GPU hours across all experiments (7 models $\times$ 3 tasks $\times$ 3 seeds): approximately 120 hours.

LLM inference was conducted via API calls (OpenAI, Anthropic, Together.ai) with approximate costs:

- GPT-4o: \$12.40 (few-shot prompting across all tasks)
- Claude-3-Haiku: \$2.80 (zero-shot prompting)
- LLaMA-3-70B, Qwen-2.5-72B: \$8.60 (Together.ai, zero/few-shot)
- Nemotron-Super: \$3.20 (zero-shot)

Total experimental cost: approximately \$27.00 for LLM inference.

Appendix-13: Additional Ethical Safeguards, Dataset Governance, and Release Considerations

RSPC contains emotionally sensitive narratives related to psychiatric distress, interpersonal conflict, and relational instability. Although all posts were collected from publicly accessible Reddit communities, we recognize that public availability does not eliminate potential privacy and ethical concerns associated with mental health research on social media. Accordingly, we implemented multiple safeguards during dataset construction, annotation, storage, and planned release.

A. Privacy Preservation and De-identification

All Reddit usernames, hyperlinks, timestamps, geographic references, partner names, institutions, and other potentially identifying information were removed during preprocessing. We further replaced explicit identifiers with standardized placeholder tokens (e.g., [USER], [PLACE], [DATE]). Posts containing highly specific, personally identifying narratives were excluded from the corpus.

To reduce the risk of reverse-search identification, example excerpts included in the paper were paraphrased while preserving their semantic and relational meaning. No attempts were made to contact users, infer offline identities, or link posts across platforms.

B. Clinical Interpretation and Annotation Constraints

Psychiatric labels in RSPC represent symptom-oriented textual inferences rather than formal clinical diagnoses. Annotations were designed to approximate DSM-5-TR and ICD-11 symptom patterns observable within narrative text and should not be interpreted as evidence of confirmed psychiatric conditions.

Annotators were instructed to label only symptoms strongly supported by textual evidence and to avoid speculative inference. Ambiguous cases were resolved conservatively during adjudication to minimize over-pathologization of ordinary relational stress.

C. Annotator Well-Being Protections

Because the dataset contains emotionally distressing content involving anxiety, abandonment fears, depressive ideation, loneliness, and interpersonal conflict, annotator well-being protocols were incor-

porated throughout the annotation process. These safeguards included:

- optional annotation breaks,
- rotating annotation schedules,
- capped daily exposure limits,
- collaborative adjudication rather than isolated review, and
- Access to mental health support resources if required.

Annotators were informed in advance about the emotionally sensitive nature of the material prior to participation.

D. Dataset Governance and Controlled Release

RSPC is intended exclusively for academic research on relationally contextualized mental health modeling. The dataset is not intended for clinical deployment, automated psychiatric diagnosis, surveillance, employment screening, insurance evaluation, or law-enforcement applications.

To reduce misuse risks, dataset release will follow controlled-access procedures:

- Researchers must agree to a non-commercial research-use license,
- Redistribution of raw data will be prohibited,
- Attempts to re-identify users will be explicitly forbidden,
- Users must acknowledge the limitations of psychiatric inference from social media text, and
- Derivative systems intended for high-stakes decision-making will not be permitted under the release agreement.

Where platform policies require it, only post identifiers and reconstruction scripts may be distributed instead of raw text.

E. Biases and Representational Limitations

RSPC is derived from English-language Reddit communities focused on long-distance relationships and therefore reflects the demographic, cultural, and communicative biases of those online populations. The benchmark may underrepresent:

- non-English speakers,

- older populations,
- offline relationship experiences,
- culturally specific relationship norms, and
- Individuals without access to digital support communities.

Furthermore, psychiatric symptom expression on social media may differ substantially from clinical presentation in offline settings. Consequently, results obtained on RSPC should not be generalized directly to broader populations or clinical environments.

F. Intended Research Scope

RSPC is designed to support research on:

- relationally grounded mental health modeling,
- contextual psychiatric symptom inference,
- socially situated affect analysis,
- interpretable computational psychiatry, and
- Longitudinal and relational reasoning in NLP systems.

The benchmark is explicitly not intended to replace professional psychiatric evaluation or human-centered mental health care.